EVIDENCE DOSSIER · TOPIC F

Public-chain proof, decoded and linked.

A receipt-level record of an adversarial exam commitment, evidence-bound qualification, delayed payment request, and successful Tier-2 execution on Base Sepolia.

REAL PUBLIC TESTNET

CONTROLLED · MOCKUSDC

Network
Base Sepolia · chain 84532

Evidence freeze
19 October 2026 UTC

Scope
Research prototype · no customer funds

01 · VERIFICATION STATEMENT

What this dossier proves—and what it does not

Four public testnet transactions establish a coherent path from committed evaluation evidence to a contract-enforced payment. Every asserted successful action has a status-success receipt and a matching decoded event.

Proven

- Contract bytecode existed at the recorded vault address.
- The examiner committed the configuration and corpus hashes.
- The owner activated a qualification for the exact agent key.
- The agent created pending request `2`.
- The agent later called `executePending(2)`.
- The receipt succeeded and emitted a matching Tier-2 `PaymentExecuted` event.

Not claimed

- No mainnet or real-funds activity.
- No external audit or certification.
- No production custody claim.
- No live Tier-1 authority: the current qualification grants `t1=0`.
- No claim that a testnet demonstration predicts operational performance.

Classification: CONTROLLED PUBLIC TESTNET. Token amounts are 6-decimal MockUSDC base units. Private keys never appear in this dossier, repository, or transaction notes.

Verification rule

An action is marked executed only when the RPC receipt reports success and the expected event decodes with matching contract, recipient, amount, invoice key, and tier. The canonical machine-readable companion is `evidence/onchain/EVIDENCE_MANIFEST.json`.

02 · DEPLOYMENT PROVENANCE

Contracts and creation transactions

COMPONENT	ADDRESS	CREATION TRANSACTION	SOURCE STATUS
MockUSDC 6-decimal test token	0x2e71d44170F788E892b4283c0a179bae8B8936F3	0x2289b55f808e430a6449359ab0bc1cb0c9f3340342bdd56abf9e239885beb7f6	Blockscout verified
BondPool Collateral and claims	0xB97Be73323731209e183Bc02a0F5d23e36Cc5902	0x645ab04975f1ea23ef66ac784b1e8d83f8efd29ed9b71c105f0a99041213f320	Blockscout verified
UnderwriteVault Funds and policy enforcement	0x8a1B3BCA25B47A0f375a649b12494a1E011eD1B0	0xf138bed8bc5cbc3c95e9bbbe2cbf908ed6aec814f9a0fa67e618e34c94433094	Sourcify exact match

0.8.28
SOLIDITY COMPILER

1.0.0
VAULT VERSION

84532
CHAIN ID

Verification boundary. Source publication makes deployed bytecode reproducible and inspectable; it is not evidence of a third-party security audit.

Public roles used in the controlled run

Owner	0x04Be79C38792ccC922bA9CC757D61B1801e9e373	Activates qualifications; may veto or revoke.
Examiner	0xe25de67dEc39AB5D7f894D963FF73B48e664F6DB	Commits exam evidence and signs the attestation.
Agent	0x0Fb2a52212Dc88CE6c29f1Ee91042ACB3eaE0B16	Requests and executes only contract-permitted payments.

03 · COHERENT EXECUTION PATH

Evidence → authority → delayed execution

08:40:00 UTC · BLOCK 48,315,456

1. ExamCommitted

Examiner anchors `configHash` and `corpusHash`; the contract assigns commit ID **5**.

08:40:02 UTC · BLOCK 48,315,457

2. QualificationActivated

Owner activates qualification `0x303f…bb4a2` for the exact agent key. Result: **T1 = 0**; T2 permitted up to 2,000 MockUSDC per request.

08:40:04 UTC · BLOCK 48,315,458

3. PaymentRequested

Agent requests 35 MockUSDC to a controlled allowlisted recipient. The pending request ID is **2**; a 300-second owner/guardian challenge window begins.

08:45:10 UTC · BLOCK 48,315,611

4. PaymentExecuted

After the challenge window, calldata decodes to `executePending(2)`. Receipt succeeds and creates permanent payment record **1**, emitting Tier-2 `PaymentExecuted`.

> The model did not choose the address, token, amount limit, or execution tier. Deterministic code and the vault enforced them.

Why ID 2 becomes payment record 1. `nextPendingId` and `nextPaymentId` are separate counters. This is expected contract behavior, not an evidence mismatch.

04 · QUALIFICATION RECEIPTS

Commitment and activation

RECEIPT SUCCESS

ExamCommitted

0x3fd70ecf8a582df153abf99eaf900c2dad569cb3c813617329155640b6b7f58f

Event fields

configHash 0x07711e2d0f1c8fdfcb6db33d55fd19b2acc75fa4ffe36b17654c4a1d64f5342e
corpusHash 0xa0637ec69a9a1bb90cb641188b268c47a8135d6882d2fdee42a2c96320646512
commitId 5

RECEIPT SUCCESS

QualificationActivated

0x9e3bef8566dd60885a2b63c39a2c96f8d05879a101bf24ef3963267b1eef9d8c

Event fields

qualId 0x303f1ad2e95898e7d07cad084339da1d84a6455c19e7d5db86b679680e1bb4a2
agentKey 0x0Fb2a52212Dc88CE6c29f1Ee91042ACB3eaE0B16
t1 0
t2MaxPerTx 2000000000

Safety result, not a failed demo. The honest gamma report triggered the compiler's utility gate, so autonomous T1 authority was zero. UNDERWRITE preserved T2: the agent may prepare a bounded request, but execution waits through the human challenge window.

Attestation binding

The signed EIP-712 qualification binds the vault, chain, agent key, configuration hash, report hash, corpus hash, exam commit ID, owner inputs, authority limits, timestamps, and examiner. Activation additionally requires an owner transaction.

Examiner key
Evidence attestation

+

Owner key
Vault consent

→

Qualification
On-chain envelope

05 · PAYMENT RECEIPTS

Request 2, payment record 1

CONTROLLED REQUEST

PaymentRequested

`0x20a245ff8141a3be094dce092db2da67583f529b1eae713b0898bb0a1b3594f6`

Receipt	**success**
Pending ID	**2**
Recipient	`0x1111111111111111111111111111111111111111`
Amount	35,000,000 base units
Ready at	08:45:04 UTC

EXECUTED + EVENT MATCHED

PaymentExecuted

`0xf19d5e201545fbe71c83c0dbd62af34a398a0414979bdaccd30fe2de827ad712`

Receipt	**success**
Input	`executePending(2)`
Payment ID	**1**
Amount	35 MockUSDC
Tier	**2**

Cross-transaction identity

FIELD	REQUEST	EXECUTION	RESULT
Agent	`0x0Fb2a52212Dc88CE6c29f1Ee91042ACB3eaE0B16`		Match
Vault	`0x8a1B3BCA25B47A0f375a649b12494a1E011eD1B0`		Match
Pending reference	Event ID 2	Calldata ID 2	Match
Recipient	`0x1111111111111111111111111111111111111111`		Match
Amount	35,000,000	35,000,000	Match

06 · INDEPENDENT VERIFICATION GUIDE

Verify without trusting this PDF

The evidence is designed to be checked from public chain data, repository artifacts, and deterministic decoding—not from screenshots.

1. Open each transaction in a Base Sepolia explorer.
2. Confirm destination `0x8a1B…D1B0`, receipt success, and block number.
3. Decode the request as `requestPayment(address,uint256,bytes32,bytes32)`.
4. Decode the execution as `executePending(uint256)`; the argument must be **2**.
5. Decode logs against the published ABI; confirm recipient, amount, invoice key, and tier.
6. Hash the raw receipt files and compare them with the manifest.

Minimal RPC verification

```
cast receipt 0xf19d5e201545fbe71c83c0dbd62af34a398a0414979bdaccd30fe2de827ad712 --rpc-url
<BASE_SEPOLIA_RPC>

cast tx 0xf19d5e201545fbe71c83c0dbd62af34a398a0414979bdaccd30fe2de827ad712 --rpc-url
<BASE_SEPOLIA_RPC> --json

cast decode-calldata "executePending(uint256)" <INPUT>
```

Receipt integrity hashes

real-01-exam-committed.json	`4c4f9889339a0149ccbdcdcfd01f4e887991d5c5e53f266b96c3a24ea98263dc`
real-02-qualification-activated.json	`6366405ef548afb2c1d25cff034ccb7c0ec4fc5d3db8f50ef4ab771a4cd7e535`
real-03-payment-requested.json	`a3471ae8c2a766bb1bb3e1c2cb0174c8e96e019fb83f43b603aab9d171b7224d`
real-04-payment-executed.json	`b2a0533d318a2d174884511de5e2970f25516a3dfa98a9ff6bf41f9a7eb4acc0`

07 · EVIDENCE REGISTER

Explorer links and artifact map

Exam commitment

Scan or click to inspect
`0x3fd7…f58f`

Qualification

Scan or click to inspect
`0x9e3b…9d8c`

Payment request

Scan or click to inspect
`0x20a2…4f6`

Payment execution

Scan or click to inspect
`0xf19d…712`

Repository artifacts

Machine-readable manifest	`evidence/onchain/EVIDENCE_MANIFEST.json`
Human-readable log	`evidence/onchain/EXECUTION_EVIDENCE.md`
Raw receipts	`evidence/onchain/receipts/*.json`
Deployment record	`contracts/deployments/84532.json`
Contract ABI	`engine/src/chain/abi/UnderwriteVault.json`

Evidence conclusion. The recorded transaction chain demonstrates real contract enforcement on a public test network. It does not substitute for a professional audit, independent model evaluation, mainnet pilot, or production key-management review.